Empathetic Dialogue Classification

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Abstract

Using the *EmpatheticDialogues* dataset, we compare traditional machine learning methods and deep learning methods for classifying conversations into 19 emotion classes. We implement standard preprocessing techniques such as TF-IDF, as well as develop a new emotion classification scheme based on highly correlated emotions. We then develop a hierarchical model for classification that works from coarse-grain to fine-grain classification. This model performs better than any traditional machine learning method alone. We then test a BERT-based model, further improving the accuracy of the classification. To conclude, we compare these two models.

1 Introduction

Empathy is the ability to understand and share the feelings of others. It plays a crucial role in effective human communication. In recent years, with the rapid growth of digital communication platforms, there has been an increasing interest in enabling machines to recognize and respond to human emotions with empathy. This has led to the development of various computational models for emotion and empathy detection in text, particularly within the fields of Natural Language Processing (NLP) and affective computing (Calvo & D'Mello, 2010; Cambria et al., 2017).

Emotion classification from text involves analyzing linguistic features to identify the emotional state conveyed by the user. Traditional machine learning (ML) approaches, such as Support Vector Machines (SVM) and Logistic Regression, have long been used for this purpose, relying heavily on feature engineering and structured representations of text (Mohammad et al., 2013). However, with the advent of deep learning, models such as recurring neural networks (RNNs), convolutional neural networks (CNNs) and transformer-based architectures such as BERT have demonstrated superior capabilities to capture complex linguistic patterns and contextual information, thus improving classification performance (Devlin et al., 2018; Yang et al., 2019).

This report focuses on building a text-based emotion/empathy classifier and comparing the effectiveness of traditional ML techniques with deep learning models. In particular, we compare a hierarchical model using logistic regression and a BERT-based model and compare the accuracy of emotion classification using the Empathetic Dialogues dataset (Rashkin et al. 2019).

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2 Problem

Although traditional machine learning and deep learning approaches have been applied to text-based emotion analysis, it is still necessary to systematically compare their effectiveness in classifying emotional and empathetic content (Buechel & Hahn, 2017). Traditional ML methods are generally less resource-intensive and easier to interpret, but may fail to capture nuanced emotional cues in text (Mohammad et al., 2013). In contrast, deep learning models, though often requiring larger datasets and computational power, have the potential to better understand context, semantics, and subtle emotional expressions (Devlin et al., 2018; Yang et al., 2019).

The central problem addressed in this report is the following: *How do traditional machine learning techniques compare with deep learning models in building an effective text-based emotion/empathy classifier?* By conducting a comparative analysis of both approaches, this study aims to provide information on their respective strengths, limitations, and applicability for emotion and empathy detection tasks.

3 Related work and background information

Emotion and empathy classification in text have grown and evolved greatly with advancements in NLP. In this section we contextualize our work within foundational and contemporary research, with emphasis on methodologies, datasets, and gaps we aimed to address. Early emotion classification frameworks, such as Ekman's six basic emotions (Ekman, 1992) and Plutchik's wheel of emotions (Plutchik, 1980), laid the groundwork for computational models. Recent studies, such as (Rashkin et al., 2019), explored this by introducing the EmpatheticDialogues dataset, which was developed to train models to recognize and respond to nuanced emotional cues in conversations. Their work highlights the importance of context-aware empathy in dialogue systems, a focus shared in our project. Traditional ML methods, such as SVM and Logistic Regression, have been used for text classification. For instance, Mohammad et al. (2013) employed lexicons to detect emotions in tweets, achieved moderate accuracy but struggled with contextual nuances. Likewise, Wang et al. (2012) showed that SVM performs well in sentiment analysis but struggles with handling hierarchical or nuance emotions. These studies highlight the challenges we addressed: sparse text representation like TF-IDF and limited capacity to model emotional hierarchies. The coming of transformer architectures, in particular BERT (Bidirectional Encoder Representations from Transformers), revolutionized NLP by effectively capturing context from both directions (Devlin et al., 2018). BERT excels in sentiment analysis (Chi et al. 2019) and emotion detection because it is pretrained on a large corpora of data and can be fine-tuned for specific tasks. Our work builds on these findings, applying BERT for emotion classification on conversational data. Hierarchical models address emotion granularity by grouping labels into coarse-to-fine categories. Alm et al. (2005) pioneered this approach, mapping emotions into broad classes before subdividing them. Our hierarchical framework aligns with these strategies, structuring 32 emotions into positive and negative classes to mitigate class imbalance and reduce error rates. For this study we used the EmpatheticDialogues (ED) dataset (Rashkin et al. 2019). The ED data set is a benchmark for empathy-driven dialogue systems, comprising 24,850 crowdsourced conversations collected via Amazon Mechanical Turk (MTurk) from 810 participants, structured into training (78%), validation (11%), test (10%) sets. In Figure 1 each conversation involves a Speaker sharing an emotional situation and a Listener responding empathetically.

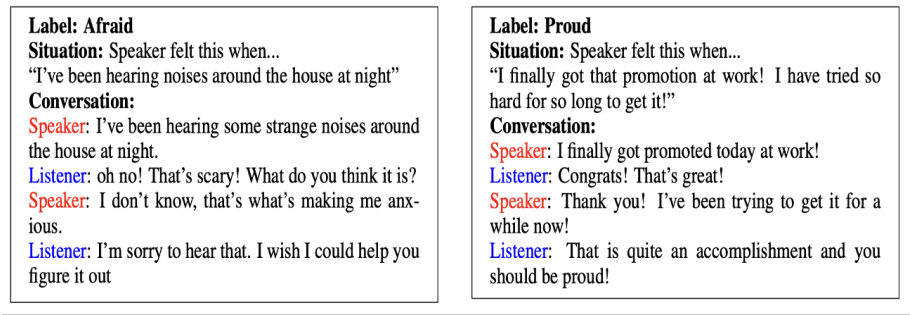


Figure 1: Diagram of an example of how EmpatheticDialogues dataset is structured (Source: Rashkin et al., 2019).

The dataset is annotated with 32 emotion labels, ensuring balanced coverage of both positive and negative emotions. Conversation average 4.31 turns, with the Speakers describing situations in 1-3 sentences (avg. 19.8 words) and Listeners responding naturally without access to the emotion label or situation description.

Emotion	Most-used speaker words	Most-used listener words	Training set emotion distrib
Surprised	got,shocked,really	that's,good,nice	5.1%
Excited	going,wait,i'm	that's,fun,like	3.8%
Angry	mad,someone,got	oh,would,that's	3.6%
Proud	got,happy,really	that's,great,good	3.5%
Sad	really,away,get	sorry,oh,hear	3.4%
Annoyed	get,work,really	that's,oh,get	3.4%
Grateful	really,thankful,i'm	that's,good,nice	3.3%
Lonely	alone,friends,i'm	i'm,sorry,that's	3.3%
Afraid	scared,i'm,night	oh,scary,that's	3.2%
Terrified	scared,night,i'm	oh,that's,would	3.2%
Guilty	bad,feel,felt	oh,that's,feel	3.2%
Impressed	really,good,got	that's,good,like	3.2%
Disgusted	gross,really,saw	oh,that's,would	3.2%
Hopeful	i'm,get,really	hope,good,that's	3.2%
Confident	going,i'm,really	good,that's,great	3.2%
Furious	mad,car,someone	oh,that's,get	3.1%
Anxious	i'm,nervous,going	oh,good,hope	3.1%
Anticipating	wait,i'm,going	sounds,good,hope	3.1%
Joyful	happy,got,i'm	that's,good,great	3.1%
Nostalgic	old,back,really	good,like,time	3.1%
Disappointed	get,really,work	oh,that's,sorry	3.1%
Prepared	ready,i'm,going	good,that's,like	3%
Jealous	friend,got,get	get,that's,oh	3%
Content	i'm,life,happy	good,that's,great	2.9%
Devastated	got,really,sad	sorry,oh,hear	2.9%
Embarrassed	day,work,got	oh,that's,i'm	2.9%
Caring	care,really,taking	that's,good,nice	2.7%
Sentimental	old,really,time	that's,oh,like	2.7%
Trusting	friend,trust,know	good,that's,like	2.6%
Ashamed	feel,bad,felt	oh,that's,i'm	2.5%
Apprehensive	i'm,nervous,really	oh,good,well	2.4%
Faithful	i'm,would,years	good,that's,like	1.9%

Figure 2: Diagram of how Empathetic Dialogues dataset is distributed among the 32 Emotions (Source: Rashkin et al., 2019).

Unlike datasets such as MELD (Poria et al., 2019), which focus on multimodal emotion recognition, or DailyDialog (Yanran et al., 2017), which emphasizes casual exchanges, EmpatheticDialogues prioritizes empathetic listener responses. Our work leverages its structure to compare a traditional ML model and BERT, addressing gaps in prior studies that overlooked cross-technique comparisons on this dataset.

4 Preprocessing

To provide the classifier with richer contextual information, the first preprocessing step involved combining speaker and listener utterances into a single sequence for each conversation. Rather than using individual utterances, all dialogue turns were combined in the order they appeared, resulting in a single text string that represented the full conversation between participants. For example, as shown in Table 1, each conversation has an associated emotion label, and the combined utterance.

This approach enabled the model to better capture the emotional context by providing a more nuanced input that simplified the classification task. Following the combined conversation utterances, the next preprocessing step involved tokenization and stopword removal. Using the `nltk` library in Python, each combined utterance was first tokenized—i.e., split into individual words to prepare the text for further processing. After tokenization, common stopwords such as "the," "is," and "and" were removed. These words occur frequently across all conversations and carry low semantic weight, so do not contribute significantly to the classification task. Hence, removing them helped reduce noise and dimensionality in the feature space. This process is shown in Figure 3.

Combined Utterance	Emotion Label
I've been hearing some strange noises around the house at night. Oh no! That's scary! What do you think it is? I don't know, that's what's making me anxious. I'm sorry to hear that. I wish I could help you figure it out.	Afraid
I finally got promoted today at work! Congrats! That's great! Thank you! I've been trying to get it for a while now! That is quite an accomplishment and you should be proud!	Proud

Table 1: Examples of combined conversations with corresponding emotion labels

The final preprocessing step involved converting the cleaned text into a numerical representation using the Term Frequency-Inverse Document Frequency (TF-IDF) technique. TF-IDF is a widely used statistical measure that reflects how important a word is to a document in a collection of

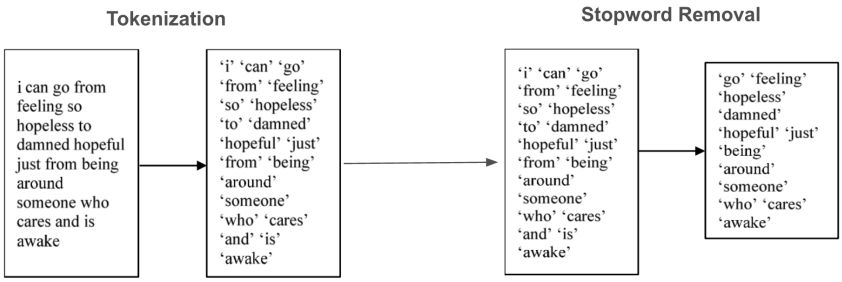


Figure 3: Tokenization followed by stopword removal in the preprocessing pipeline.

documents. It assigns higher weights to words that are frequent in a specific document but relatively rare across the corpus, helping highlight more meaningful terms for classification.

The TF-IDF score is computed as the product of two components: term frequency (TF) and inverse document frequency (IDF). The term frequency captures how often a word appears in a specific document relative to the total number of words in that document:

$$\text{TF} = \frac{\text{Number of times a word "X" appears in a Document}}{\text{Number of words present in the Document}}.$$

The inverse document frequency downweights words that are common across many documents in the corpus:

$$\text{IDF} = \log\left(\frac{\text{Number of Documents in the Corpus}}{\text{Number of Documents where word "X" has appeared}}\right).$$

Multiplying the two gives the final TF-IDF score:

$$\text{TF-IDF} = \text{TF} \times \text{IDF}.$$

This representation is important for emotion classification because it helps the model focus on the words that actually carry emotional meaning, rather than common words that appear in every conversation. By highlighting these key terms, TF-IDF improves the model's ability to understand and predict the emotion behind each conversation.

5 Traditional Models

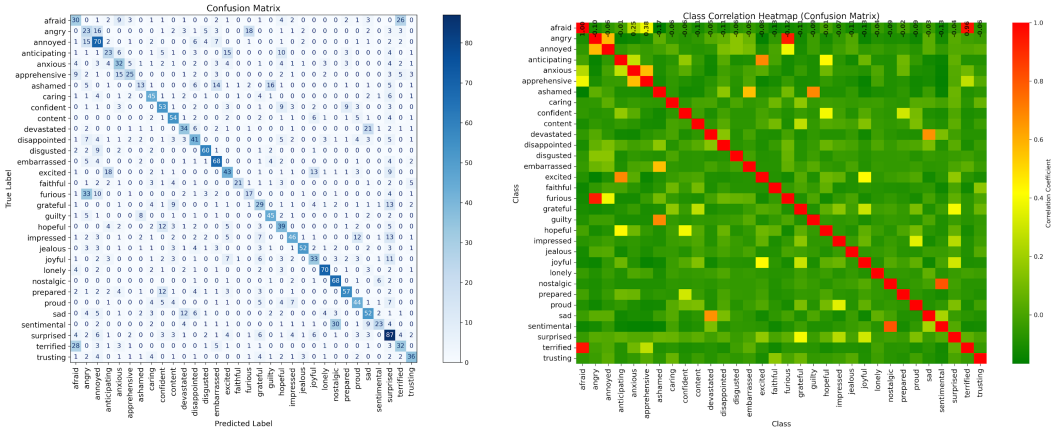
After applying the initial preprocessing steps, we experimented with several traditional classification models to evaluate their effectiveness on the task of emotion classification. These models included Linear Support Vector Machines (SVM), Logistic Regression, Random Forest, Naive Bayes, and Decision Trees. For each model, we conducted hyperparameter tuning to optimize their performance. Table 2 presents the training and validation error for each model. As seen, while some models like Logistic Regression and Random Forest performed better, the overall validation errors remained high, indicating that the models did not generalize well to unseen data.

Model	Training Error	Validation Error
Linear SVM	0.7986	0.8125
Logistic Regression	0.3222	0.5548
Random Forest	0.3868	0.6609
Naive Bayes	0.1125	0.5802
Decision Tree	0.0	0.7505

Table 2: Training and validation errors for traditional models.

To better understand the limitations in model performance, we performed a detailed error analysis using a confusion matrix, focused on the SVM model's predictions on the validation set. As shown in Figure 4a, the confusion matrix revealed frequent misclassifications between semantically similar emotions. For instance, emotions such as *angry*, *annoyed*, and *furious* were often confused with one another,

suggesting a strong overlap in their linguistic expressions. To further explore these patterns, we generated a correlation heatmap (Figure 4b) that visualized the relationships between emotion labels. High correlation values between pairs such as *afraid* and *terrified* reinforced the intuition that certain categories were closely aligned. These insights reflected the inherent challenges of fine-grained emotion classification, where even humans may struggle to distinguish between highly similar emotional states.



(a) Confusion Matrix

(b) Correlation Heatmap

Figure 4: Classification Report for SVM on the validation set

Based on these insights, we decided to combine certain emotion categories to simplify the classification task. Initially, we worked with 32 emotion labels, but after testing with traditional models and analyzing the confusion matrix and heatmap, we reduced the number of labels to 19, as shown in the emotion mapping scheme (Table 3).

6 Hierarchical Model

The hierarchical machine learning model that we designed to classify text into 19 distinct emotion classes utilizes a classic machine learning method integrated into a unique architecture. By breaking down the classification task into three steps, starting with a coarse classification that gets progressively more specific, this model performs better than a single machine learning method alone. The model begins by constructing a hierarchical classification system for the 19 distinct emotion labels. The breakdown of emotion classification is described in Figure 11. With this classification system, we can train a hierarchical model.

After standard preprocessing, as described in Section 4, each utterance is vectorized using TF-IDF (described in Section 4). The hierarchical model can best be described as the composition of a decision tree and logistic regression. At the first level of decision, we relabel all training examples as 'positive' or 'negative' according to hierarchical emotion mapping. A logistic regression model is trained for the positive/negative classification task. In the next level, the training examples that were labeled 'positive' are relabeled as 'happy' or 'reflective' according to the

Emotion	Relabeled
afraid terrified	fear
angry annoyed furious	anger
anxious apprehensive	worried
ashamed embarrassed guilty	shame
hopeful confident prepared	hope
sad devastated	sadness
excited anticipating joyful	happy
impressed proud	recognition
nostalgic sentimental	nostalgia
jealous	jealous
grateful	grateful
surprised	surprised
caring	caring
lonely	lonely
faithful	faithful
disappointed	disappointed
disgusted	disgusted
content	content
trusting	trusting

Table 3: Emotion Mapping Scheme showing the relabeling of emotion categories.

hierarchical emotion mapping. A new logistic regression model is trained on only the 'positive' examples and labels them as 'happy' or 'reflective'. Similarly, a new logistic regression model is trained on only the 'negative' examples and labels them as 'sad' or 'angry'. At the final level of classification, we train logistic regression models on each of the 'happy', 'reflective', 'sad', and 'angry' subsets, respectively. The labels at this stage are one of the 19 emotion classes. Figure 5 illustrates the architecture of the hierarchical model.

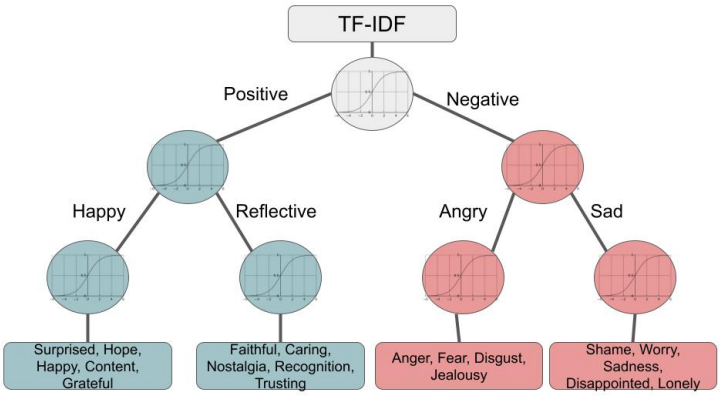


Figure 5: Diagram of the architecture for the hierarchical model. Level 1 is the positive/negative classification. Level 2 is the happy/reflective or angry/sad classification. Level 3 is the final classification as one of the 19 emotion classes. Each node is a distinct logistic regression model that is trained only on the examples labeled as the input, and classifies each example as exactly one of the output labels.

Using 5-fold validation, we tuned the regularization parameter C for each level of the hierarchical model. As the number of classes increases, the tuned regularization parameter increases. This results in less regularization as the classification becomes finer (note that for logistic regression in scikit-learn, smaller C values induce stronger regularization). This positive correlation between C values and the level of the hierarchical model is to be expected since emotion classes become more similar as we move through the model.

After the model has been trained, we test the model on the test set. Again, each test utterance is vectorized using TF-IDF. Each test example is then filtered through the decision tree until it reaches a leaf and gets classified as one of the 19 examples. Figure 12 shows the evolution of the confusion matrices through the hierarchical model for classifying 'shame', 'worry', 'sadness', 'disappointed', and 'lonely'. A limitation of this model is that any misclassifications in level 1 or 2 prevents the example from being correctly classified in the end. It is crucial to minimize the errors in level 1 and 2 as much as possible to reduce the impact of misclassifications early on.

Figure 14 shows the change in error for each level of the model. As the number of classes increases, the error increases, as expected. The training error decreases as the test error increases, indicating overfitting. However, the final error for the hierarchical model on the test set with a 95% confidence interval is 0.3954 ± 0.0190. This not only does better than the random classification error of $.9474$, but it is also an improvement on the traditional machine learning methods on their own.

7 Deep Learning Approach

The validation error was still high after experimenting with traditional machine learning algorithms and hierarchical models. The emotional mapping as defined in Table 3 to combine similar emotions improved the classification accuracy to some extent, but the results were still unsatisfactory. This could be due to the reason that traditional word embeddings, such as TF-IDF, do not understand words contextually. There are some other embedding techniques, i.e. GloVe and Word2Vec, but they assign one meaning to a word regardless of the context of the whole sentence. Given the complexity of the dataset, accurately classifying emotions from input conversations required the use of advanced contextual word embeddings typically provided by Large Language Models (LLMs) in deep learning.

7.1 BERT - Bidirectional Encoder Representations from Transformers

BERT is a deep learning model developed by Google in 2018 (BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding, 2019), that excels in natural language understanding (NLU) tasks like text classification, question answering, and text generation.

7.1.1 Key Features

Unlike older NLP models that process text from left to right (e.g., GPT) or right to left, BERT reads entire sentences at once, learning context from both directions. BERT is pre-trained on large datasets (Wikipedia, BooksCorpus), and then fine-tuned for specific tasks like sentiment analysis and text classification. Traditional models like Word2Vec assign a single meaning to a word, but BERT dynamically understands words based on context (e.g., "bank" in "riverbank" versus "money bank"). It relies on the Transformer architecture, specifically the encoder component, using self-attention mechanisms to determine the importance of each word in relation to others when making predictions.

7.1.2 BERT Base

Google originally developed two versions of BERT (Base and Large). Due to computational limitations, we used BERT Base for the emotional classification of our dataset. The BERT Base architecture is given in Figure 6.

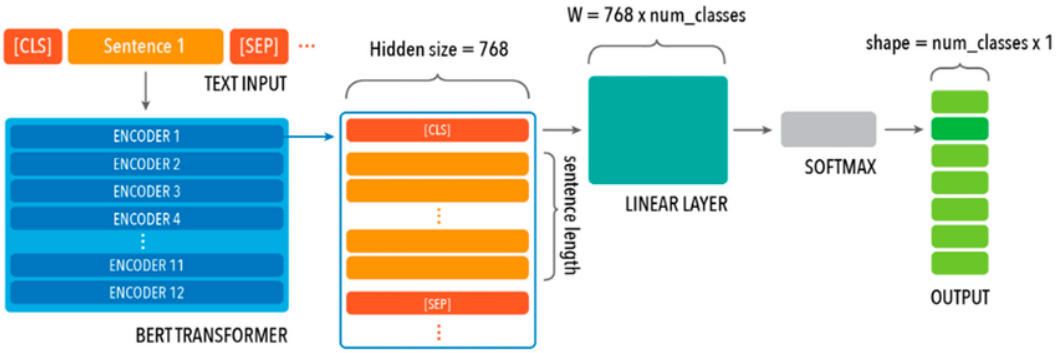


Figure 6: BERT Base Uncased Model Architecture (Source: Muller, B. (2022))

BERT is a complicated model and there are a large number of hyperparameters that can be tuned to achieve the desired results. Some of the major components of model training are given in Table 4. A detailed overview of our BERT model parameters can be found in Appendix B.

Table 4: Key Model Components - BERT

Component	**Purpose**
AdamW Optimizer	Optimizes BERT while reducing weight decay bias
CrossEntropyLoss	Computes classification loss
Mixed Precision (FP16)	Speeds up training & reduces memory usage
Gradient Accumulation	Allows large batch training on limited memory
Cosine Annealing LR Scheduler	Smoothly adjusts learning rate to prevent sharp changes
Validation	Evaluates model performance after each epoch

7.1.3 Results - Training vs Validation Set

It is important to note that the emotion mapping in Table 3 was applied to training, validation, and test sets before training and evaluation of the BERT model. After multiple iterations on the training and validation set with different model parameters, we were able to achieve the validation error of 24% (76% accuracy), which is a significant improvement compared to traditional ML models. The plot of training and validation error against the number of epochs is shown in Figure 7. which clearly shows that after the third epoch, the validation error plateaus and does not improve while the training error

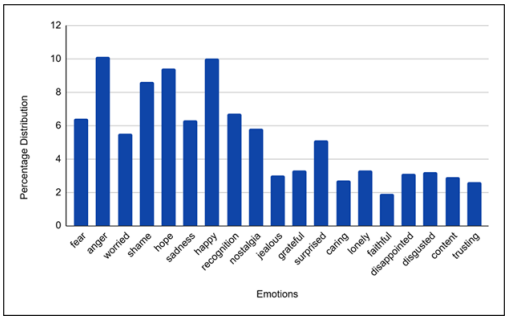
continues to drop, indicating that the model starts to overfit on the training dataset. The code was designed in such a way that it automatically saves the best trained model with the lowest validation error.



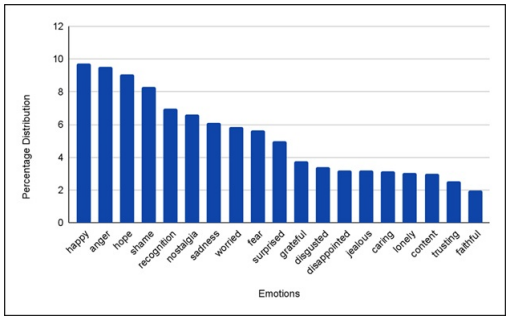
Figure 7: Training & Validation Errors vs Epochs

7.1.4 Results - Test Set

Before applying saved model with the lowest validation error, we checked the emotion class distribution in test set to make sure that the distribution was balanced between training and test sets. Figure 8a and Figure 8b show the emotion class distribution in training and test sets respectively. The emotion classes (happy, anger, hope) have the highest percentages, and (faithful, trusting) have the lowest percentages.



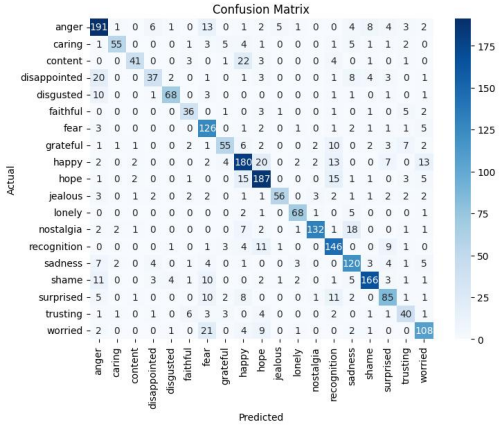
(a) Training Set Emotion Class Distribution



(b) Test Set Emotion Class Distribution

Figure 8: Emotion Class Distributions

On the test set, the misclassifications error was 25% (75% Accuracy) which is close to the model performance on the validation set. The confusion matrix and classification report are shown in Figure 9a and Figure 9b respectively.



(a) Test Set Confusion Matrix

Emotion	precision	recall	f1-score	support
anger	0.73	0.79	0.76	242
caring	0.89	0.69	0.77	80
content	0.84	0.54	0.66	76
disappointed	0.69	0.46	0.55	81
disgusted	0.88	0.79	0.83	86
faithful	0.68	0.72	0.7	50
fear	0.63	0.88	0.73	144
grateful	0.74	0.58	0.65	95
happy	0.7	0.73	0.71	247
hope	0.74	0.81	0.77	231
jealous	0.88	0.69	0.77	81
lonely	0.86	0.87	0.87	78
nostalgia	0.94	0.79	0.85	168
recognition	0.7	0.82	0.75	177
sadness	0.69	0.77	0.73	155
shame	0.87	0.79	0.83	211
surprised	0.69	0.67	0.68	127
trusting	0.58	0.63	0.6	64
worried	0.72	0.72	0.72	149
accuracy	**0.75**	**0.75**	**0.75**	**0.75**
weighted avg	**0.76**	**0.75**	**0.75**	**2542**

(b) Test Set Classification Report

Figure 9: Test Set Classification Report

The BERT model achieved an overall accuracy of 75% on a 19-class emotion classification task, with a weighted F1-score of 0.75, indicating strong general performance. We are emphasizing F1-score, precision, and recall due to the class imbalance and multi-class nature of the problem. Among the best-performing classes were lonely (F1: 0.87), nostalgia (F1: 0.85), shame (F1: 0.83), and disgusted (F1: 0.83), all demonstrating high precision and recall. These emotions were also clearly distinguishable in the confusion matrix, showing minimal confusion with other categories. On the other hand, emotions such as disappointed (F1: 0.55), trusting (F1: 0.60), and grateful (F1: 0.65) exhibited lower performance, primarily due to lower recall, suggesting difficulty in correctly identifying these classes. The confusion matrix revealed that some emotions, particularly happy, hope, and anger, were often confused with semantically similar ones, highlighting the subtle overlap between emotional expressions. Despite the presence of class imbalance, the model managed to maintain consistent performance across most categories.

Example 1:
Conversation: I got attacked by a swarm of bees a few months back. Needless to say it hurt. How awful, besides painful, it must have been **scary**! It was, they we're absolutely **furious** with me for some reason. Were you messing with them, maybe? or taking their honey?
True Label: anger
Predicted Label: fear

Example 2:
Conversation: I have to rely on my prayers and my belief in a high power to get me through not having a job and no money. Things work out. It doesn't seem like it but they do. I remind myself all the time that it will work out and all will be good again. It will. I am sure of it.
True Label: faithful
Predicted Label: trusting

Example 3:
Conversation: My husband was teaching me how to make a steak. The flame was so warm! That sounds like a fun thing to do as a couple. It was but I was so sad that I was so **scared** Why were you so **scared**?
True Label: worried
Predicted Label: fear

Figure 10: Misclassified Examples - Test Set

Figure 10 shows some of the randomly chosen misclassified examples from the test set. These examples indicate the close relation between true and predicted labels. In the first example, the words "scary" and "furious" are both present which might have confused the classifier to correctly predict the label as these words are associated with fear (predicted label) and anger (true label) respectively. In the second example, the classifier recognized a core emotion (trusting/faithful) but couldn't distinguish between spiritual faith and general trust due to subtle textual cues. In the third example, the emotion "worried" is on the spectrum of "fear", and this subtlety is often missed in classification models. These examples demonstrate that while the predictions are not exact, they

are still emotionally reasonable. Misclassifications like these often result from label ambiguity, the semantic similarity between emotions, and limitations in dataset annotation rather than model failure. The model tends to generalize based on dominant cues in the input text. If a conversation hints at multiple emotions, the model may pick the most statistically likely one, even if it's not the true label.

8 Conclusion

This project aimed to compare the effectiveness of traditional machine learning techniques and deep learning models in building a text-based emotion/empathy classifier. Through our experiments and evaluation, we found that deep learning models, particularly BERT, significantly outperformed traditional ML approaches in terms of accuracy and F1-score. BERT's ability to automatically learn contextual and semantic features from raw text made it more effective for capturing nuanced emotional expressions, whereas traditional models relied heavily on manual feature engineering such as TF-IDF.

However, this improved performance comes at a cost. Deep learning models require high computational resources, longer training times, and access to GPUs, making them less suitable for low-resource environments or time-sensitive applications. However, traditional ML models offer faster training times and can run efficiently on standard CPUs, making them more practical for simpler or resource-constrained use cases.

In conclusion, the choice between traditional ML and deep learning should depend on the specific requirements and constraints of the task. For high-stakes or complex NLP tasks where accuracy and depth of understanding are crucial, deep learning models like BERT are the preferred choice. For quick deployments in limited-resource settings, traditional ML remains a viable and efficient option.

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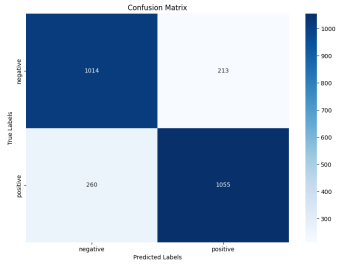
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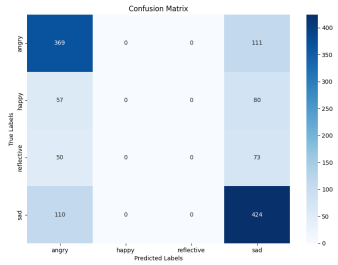
A Hierarchical model figures

Level 1	Level 2	Level 3
Positive	Happy	Surprised
		Hope
		Happy
		Content
		Grateful
	Reflective	Faithful
		Caring
		Nostalgia
		Recognition
		Trusting
Negative	Sad	Shame
		Worried
		Sadness
		Disappointed
		Lonely
	Angry	Anger
		Fear
		Disgust
		Jealousy

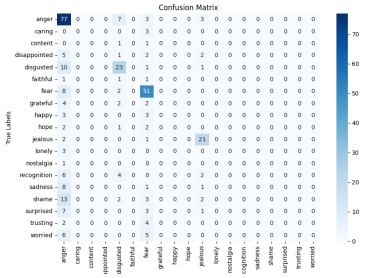
Figure 11: The hierarchical emotion mapping.



(a) Confusion matrix for the test set after level 1

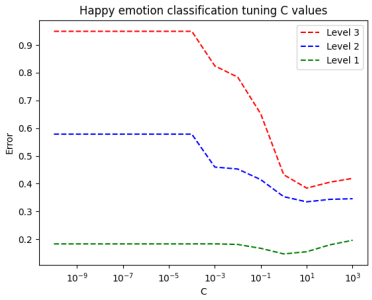


(b) Confusion matrix for the test set after level 2 classification of angry/sad.

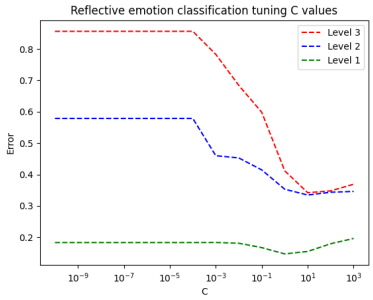


(c) Confusion matrix for the sad subset of the test set after level 3

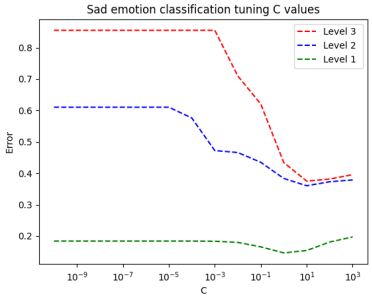
Figure 12: Confusion matrices for the hierarchical model following the 'sad' path.



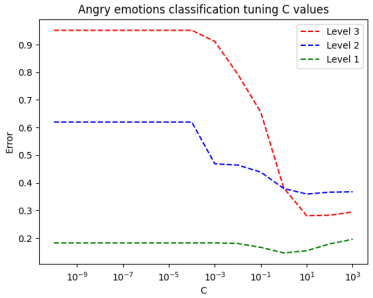
(a) C hyperparameter tuning for the 'happy' path.



(b) C hyperparameter tuning for the 'reflective' path.



(c) C hyperparameter tuning for the 'sad' path.



(d) C hyperparameter tuning for the 'angry' path.

Figure 13: Each plot graphs the regularization parameter's effect on the error for each path of the hierarchical model. Level 1 is the positive/negative classification. Level 2 is the happy/reflective classification for Figures 13a and 13b, and the angry/sad classification for Figures 13c and 13d. Level 3 is the final round of classification for the respective subsets of the 19 emotion classes, dependent on the level 2 classification.



Figure 14: Error as the number of classes increases.

B BERT Emotion Classifier Model Parameters

Table 5: BERT Emotion Classifier Summary Table

Component	**Purpose**	**Remarks**	**Parameter Values**
Tokenizer	Tokenize input text	Converts raw text to model-readable token IDs	"bert-base-uncased"
Dataset	Custom dataset	Tokenizes input and aligns with labels	max_length=128, padding="max_length"
Model Base	Pre-trained BERT backbone	Uses BERT to generate embeddings	"bert-base-uncased"
Dropout Layer	Prevent overfitting	Randomly zeroes out 50% of hidden units	p=0.5
Linear Layer	Classification layer	Predicts emotion class from [CLS] token	in_features=768, out_features=num_classes
Mixed Precision	Speed up training	Enables FP16 computation	Enabled
Criterion	Compute classification loss	For multi-class classification	CrossEntropyLoss()
Optimizer	Update model parameters	Weight decay improves generalization	lr=2e-5, eps=1e-8, weight_decay=0.01
Scheduler	Adjust learning rate	Smooth decay using cosine schedule	T_max=5, eta_min=1e-6
Batch Size	Training batch size	Number of samples per batch	batch_size=16
Device	Hardware accelerator	Uses GPU if available	"cuda" or "cpu"
Gradient Scaler	Stabilize FP16 training	Prevents gradient underflow	Enabled
Accumulation	Gradient accumulation	Useful with small memory GPUs	4
Epochs	Number of passes over data	Controls training duration	5
Model Save	Save best model	Saves model with lowest validation loss	"bert_emotion_classifier_best.pth"
Plotting	Visualize error trends	Plots and saves training error curves	"Training & Validation Errors vs Epochs.pdf"